Dr. Ebin Deni Raj

Associate Dean, Academic Affairs | Assistant Professor (Grade-I), CSE

Indian Institute of Information Technology Kottayam (Institution of National Importance)

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Research Metrics Summary

Total Pubs.	h-index	i10-index	Total Citations	Cumul. Impact Factor	Best Papers	Funded Projects
53+	13	17	1139+	39.355	9	6+

Source: Google Scholar (Feb 2026) | ACM Distinguished Speaker | IEEE Senior Member | TEDx Speaker

Education

Degree	Year	Institution	Specialisation / Thesis
Ph.D.	2016	Vellore Institute of Technology (VIT), Vellore	Effective Strategies for Information Spreading in Online Social Networks
M.Tech.	2012	Vellore Institute of Technology (VIT), Vellore	Information Technology & Engineering
B.Tech.	2010	Cochin University of Science & Technology (CUSAT), Kochi	Computer Science & Engineering

Professional Experience

Position	Organisation	Period
Associate Dean, Academic Affairs	IIIT Kottayam	2022 – Present
Head of Department, CSE	IIIT Kottayam	(Prior Tenure)
Assistant Professor (Grade-I), CSE	IIIT Kottayam	Jan 2017 – Present
Teaching & Research Associate, SITE	VIT University, Vellore	Jul 2013 – Jun 2016
Assistant Professor, CSE	UKF College of Engg. & Tech., Paripally	Jun 2016 – Jan 2017
Assistant Professor, CSE	LBS Inst. of Technology for Women, TVM	Dec 2012 – Jul 2013
Lecturer – Cyber Law	ACE Academy, Trivandrum	Jun 2012 – Nov 2012
Project Intern	Alcatel-Lucent (Bell Labs), USA	Sep 2011 – May 2012

Also selected by Cognizant Technology Solutions (CTS) in campus placement (2011).

Courses Taught

PhD Level, CSE, CSH and ECE , IIIT Kottayam)

Code	Course Name	Level	Approx. Period
ICS801	Mathematical and Statistcial Foundtaions of Data Science	PhD	2019
ICS816	Responsible and Explianable AI	PhD	2025- Present
ICS809	Federated Learning and Explainability	PhD	2021–2023

Postgraduate Level (M.Tech. – AI & Data Science, IIIT Kottayam)

Code	Course Name	Level	Approx. Period
DSC513	Introduction to AI & Data Science	PG	2019–Present
–	R Programming	PG	2020–2023 t
–	Data Structures & Programming	PG	2017–Present

Undergraduate Level (B.Tech. CSE, IIIT Kottayam)

Code	Course Name	Level	Approx. Period
ICS111	Programming in C	UG	2017–2020
ICS213	Python Programming	UG	2019–Present
ICS112	Bash Scripting	UG	2017–2020
ICS121	Data Structures	UG	2017–2020
ICS221	Theory of Computation	UG	2017–2020
ICS224	Algorithm Analysis & Design	UG	2017–2020
ICS313	Data Warehousing & Mining	UG	2017–2020
ICS311	Parallel Algorithms	UG	2017–2020
ICS321	Advanced Data Mining	UG	2017–2020

Other Institutions

Embedded Systems; Sustainable Engineering (UKF College of Engineering & Technology, 2016–17). Cyber Law (ACE Academy, Trivandrum, 2012).

Sponsored Research Projects

Project Title	Role	Funding Agency	Amount	Year
Smartphone Imaging-based AI for Early Detection of Ocular Surface & Eyelid Tumors: Multicentric Study (AIIMS New Delhi, Kalyani, Nagpur, Bilaspur)	PI	ICMR	Rs 1.30 Cr	2026
UAV-Enabled Ultra-Reliable Low Latency Communication for Energy Efficiency & Latency Reduction in 6G Networks	Co-PI	UKIERI	Rs 60 L	2024
Responsible Computing	PI	Mozilla Foundation & USAID	Rs 25 L	2023
AI for Social Assessment (AIFORA) – collaboration with Johannes Gutenberg Univ., Mainz, Germany	PI	International	Rs 24 L	2020

Microsoft AI for Earth – AI for Detection & Classification of Diseases in Rice & Wheat	PI	Microsoft	Rs L	12.71	2019
Doctoral Fellowship	PI	TIH, IIT Palakkad	Rs L	21.84	2021
AI Automation Project	PI	Orzico	Rs 1 L		2019

Consultancy Projects

Description	Client	Amount
Development of E-learning Content (Invoice IIIT/Consultancy/33/2022-23)	Manorama Horizon	Rs 74,500
Development of E-learning Content (Invoice IIIT/Consultancy/39/2022-23)	Manorama Horizon	Rs 18,900
Development of E-learning Content (Invoice IIIT/Consultancy/40/2022-23)	Manorama Horizon	Rs 63,000
Development of E-learning Content (Invoice IIIT/Consultancy/44/2022-23)	Manorama Horizon	Rs 75,600

Patents

Indian Utility Patent Filed: "Autonomous Device for Harvesting Fruits" – Application No. 201941028625 (filed September 2020). Outcome of the AI Automation project with Orzico.
Indian Utility Patent Filed: "Real-Time Gaze Detection System on a Film AND/ORVideo Content and Method Thereof" – Application No. 202541068109 (filed July 2025).

Publications

*Total: 53+ | h-index: 10 | i10-index: 10 | Citations: 1057+ | Cumulative Impact Factor: 39.355 | * corresponding/lead author*

Journals – SCI/SCIE Indexed

J1. Jo A.A., **Raj E.D.***, Sahoo J. "Efficiency and Performance Optimization in Large Language Models through IB Fine-Tuning." *ACM Transactions on Intelligent Systems and Technology*, Vol. 16, 2025.

J2. S A., A A., **Raj E.D.***, S K. "Swin-RSIC: Remote Sensing Image Classification using a Modified Swin Transformer with Explainability." *Earth Science Informatics*, Vol. 18, 2025.

J3. Nair A.K., Coleri S., Sahoo J., Cenkeramaddi L.R., **Raj E.D.*** "Incentivized Federated Learning: A Survey." *IEEE Transactions on Emerging Topics in Computational Intelligence*, Vol. 9, 2025, pp. 3190–3209.

J4. Thomas J., **Raj E.D.*** "Improved Image Dehazing Model with Color Correction Transform-Based Dark Channel Prior." *The Visual Computer*, Vol. 40, 2024, pp. 8767–8790.

J5. Haridasan A., Thomas J., **Raj E.D.*** "Deep Learning System for Paddy Plant Disease Detection and Classification." *Environmental Monitoring and Assessment*, Vol. 195, No. 1, 2023.

J6. Nair A.K., **Raj E.D.***, Sahoo J. "A Robust Analysis of Adversarial Attacks on Federated Learning Environments." *Computer Standards and Interfaces*, Vol. 86, 2023. **[IF: 3.6]**

J7. Nair A.K., Sahoo J., **Raj E.D.*** "Privacy Preserving Federated Learning Framework for IoMT Based Big Data Analysis using Edge Computing." *Computer Standards and Interfaces*, Vol. 86, 2023. **[IF: 3.6]**

J8. **Raj E.D.***, Manogaran G., Srivastava G., Wu Y. "Information Granulation-Based Community Detection for Social Networks." *IEEE Transactions on Computational Social Systems*, Vol. 8, 2021, pp. 122–133. **[IF: 4.747]**

J9. Huifeng W., Kadry S.N., **Raj E.D.*** "Continuous Health Monitoring of Sportsperson using IoT Devices Based Wearable Technology." *Computer Communications*, Vol. 160, 2020, pp. 588–595. **[IF: 5.047]**

J10. Sharon Moses J., Nirmala M., **Raj E.D.*** "A Security Ensemble Framework for Securing the File in Cloud Computing Environments." *International Journal of Computational Science and Engineering*, Vol. 18, 2019, pp. 413–423.

J11. **Raj E.D.***, Dhinesh Babu L.D. "A Firefly Inspired Game Dissemination and QoS-Based Priority Pricing Strategy for Online Social Network Games." *International Journal of Bio-Inspired Computation*, Vol. 11, 2018, pp. 202–217. **[IF: 3.295]**

J12. Dhinesh Babu L.D., **Raj E.D.*** "Flocking Based Evolutionary Computation Strategy for Measuring Centrality of Online Social Networks." *Applied Soft Computing*, Vol. 58, 2017, pp. 495–516. **[IF: 8.263]**

J13. **Raj E.D.***, Babu L.D.D. "Effective Detection of Modular and Granular Overlaps in Online Social Networks Using Fuzzy ART." *International Journal of Fuzzy Systems*, Vol. 19, No. 4, 2017, pp. 1077–1092. **[IF: 4.085]**

J14. **Raj E.D.***, Babu L.D.D. "An Enhanced Trust Prediction Strategy for Online Social Networks using Probabilistic Reputation Features." *Neurocomputing*, Vol. 219, 2017, pp. 412–421. **[IF: 5.779]**

J15. **Raj E.D.***, Babu L.D.D. "A Fuzzy Adaptive Resonance Theory Inspired Overlapping Community Detection Method for Online Social Networks." *Knowledge-Based Systems*, Vol. 113, 2016, pp. 75–87. **[IF: 8.139]**

Journals – Scopus Indexed

S1. Sasikumar M., **Raj E.D.*** "Investigating Explainability of Deep Learning Models for Sequential Data on Stock Price Prediction." *Procedia Computer Science*, Vol. 258, 2025, pp. 4190–4201.

S2. Li C., **Raj E.D.*** "Feature Recognition of Abstract Art Painting Multilevel Based Convolutional Ancient Recognition Neural Network Method." *Journal of Interconnection Networks*, Vol. 22, 2022.

S3. Joshi V.S., Thomas J., **Raj E.D.*** "Quantized Coconut Detection Models with Edge Devices." *Journal of Interconnection Networks*, Vol. 22, 2022.

S4. **Raj E.D.***, Babu L.D.D. "A Survey on Topological Properties, Network Models and Analytical Measures in Detecting Influential Nodes in Online Social Networks." *International Journal of Web Based Communities*, Vol. 13, 2017, pp. 137–156.

S5. **Raj E.D.***, Babu L.D.D. "A Firefly Swarm Approach for Establishing New Connections in Social Networks Based on Big Data Analytics." *International Journal of Communication Networks and Distributed Systems*, Vol. 15, 2015, pp. 130–148.

S6. **Raj E.D.***, Babu L.D.D. "A Two Pass Scheduling Policy Based Resource Allocation for MapReduce." *Procedia Computer Science*, Vol. 46, 2015, pp. 627–634.

Book Chapters

B1. **Raj E.D.**, Lekha D.S., Jo A.A., Jose A.C. "Teaching Explainable Machine Learning to Interdisciplinary Learners: A Pedagogical Model for Responsible AI Education." *Communications in Computer and Information Science (CCIS)*, Vol. 2739, Springer, 2026, pp. 3–15.

B2. Konyala A.R., Jo A.A., **Raj E.D.** "Integrating Explainable AI with Machine Learning for Accurate Liver Disease Diagnosis." *Lecture Notes in Networks and Systems*, Vol. 5588 LNNS, 2025, pp. 371–395.

B3. Nair A.K., Sahoo J., **Raj E.D.** "Exploring Communication Efficient Strategies in Federated Learning Systems." *Federated Learning: Principles, Paradigms and Applications*, 2024, pp. 153–182.

B4. Nair A.K., Sahoo J., **Raj E.D.** "Analyzing Federated Learning from a Security Perspective." *Federated Learning: Principles, Paradigms and Applications*, 2024, pp. 213–236.

B5. Shahad P., **Raj E.D.** "The Role of Wireless Body Area Networks in Smart Healthcare System in the Context of Big Data and AI." *Handbook of Research on Machine Learning Enabled IoT for Smart Applications Across Industries*, 2023, pp. 185–201.

B6. Raveendran R., **Raj E.D.** "Deep Generative Models Under GAN: Variants, Applications, and Privacy Issues." *Lecture Notes in Networks and Systems*, Vol. 494, Springer, 2023, pp. 93–105.

B7. Jo A.A., Paul C.R., Arundhathi Nair S., Thomas J., **Raj E.D.** "Tabular Data Interpretation on COVID-19 Through Explainable Machine Learning." *Lecture Notes in Networks and Systems*, Vol. 730 LNNS, 2023, pp. 653–666.

B8. Thomas J., **Raj E.D.** "Effectual Single Image Dehazing with Color Correction Transform and Dark Channel Prior." *Communications in Computer and Information Science*, Vol. 1483, Springer, 2021, pp. 29–41.

B9. **Raj E.D.**, Babu L.D.D., Ariwa E. "Issues in On-Demand Cloud-Based Gaming Storage: Quality of Service and Quality of Experience." *Emerging Technologies and Applications for Cloud-Based Gaming*, IGI Global, 2016, pp. 115–139. *[Scopus indexed]*

B10. **Raj E.D.**, Babu L.D.D., Ariwa E., Nirmala M., Krishna P.V. "Forecasting the Trends in Cloud Computing and its Impact on Future IT Business." *Green Technology Applications for Enterprise and Academic Innovation*, IGI Global, 2014. *[Scopus indexed]*

Conference Papers (International, Peer-Reviewed)

C1. Shahad P., **Raj E.D.*** "Interpretability-Based Virtual Drift Detection and Adaptation Algorithm: A Case Study on Tetouan's Energy Data." *Intelligent Computing and Emerging Communication Technologies (ICEC 2024)*, 2024.

C2. Ijas H.A., Jo A.A., **Raj E.D.*** "Exploring Explainable AI in Large Language Models: Enhancing Transparency and Trust." *ICACC – Int. Conf. on Advances in Computing and Communications*, 2024.

C3. Mariyappan S., Thomas J., **Raj E.D.*** "SwinLeafExplain: Interpretable Cotton Disease Classification with SWIN Transformer." *ICACC – Int. Conf. on Advances in Computing and Communications*, 2024.

C4. Jo A.A., **Raj E.D.*** "Ethical Dimensions in Data-Driven Business: Balancing Fairness, Accountability, Transparency, and Explainability." *15th Int. Conf. on Computing Communication and Networking Technologies (ICCCNT 2024)*, 2024.

C5. Jo A.A., **Raj E.D.***, Vino A.S., Menon P.V. "Exploring Explainable AI for Enhanced Depression Prediction in Mental Health." *1st Int. Conf. on Innovations in Communications, Electrical and Computer Engineering (ICICEC 2024)*, 2024.

C6. Sasikumar M., **Raj E.D.*** "A Short Survey on Interpretable Techniques with Time Series Data." *2024 IEEE Int. Students Conf. on Electrical, Electronics and Computer Science (SCEECS 2024)*, 2024.

C7. Joseph S., Jo A.A., **Raj E.D.*** "Improving Time Series Forecasting Accuracy with Transformers: A Comprehensive Analysis with Explainability." *3rd Int. Conf. on Electrical, Electronics, Information and Communication Technologies (ICEEICT 2024)*, 2024.

C8. Ranjith R., Jo A.A., **Raj E.D.*** "Bipol-Driven Bias Analysis in Transformer Models: A Quantitative, Global, and Local Interpretability Approach for Textual Data." *IEEE 3rd World Conf. on Applied Intelligence and Computing (AIC 2024)*, 2024, pp. 304–309.

C9. Nair A.K., Sahoo J., **Raj E.D.*** "A Lightweight FL-Based UAV Detection Model using Thermal Images." *ACM Int. Conf. Proceeding Series*, 2024.

C10. Nair A.K., Sahoo J., Cenkeramaddi L.R., Jaswal G., **Raj E.D.*** "Mobile_FL: A Streamlined FL Framework for Process Optimisation via Client Clustering using Rough C-Means Algorithm." *ACM CPSS 2024 – 10th ACM Cyber Physical System Security Workshop*, 2024, pp. 88–97.

C11. Thiruthuvaraj R., Jo A.A., **Raj E.D.*** "Explainability to Business: Demystify Transformer Models with Attention-Based Explanations." *2nd Int. Conf. on Applied Artificial Intelligence and Computing (ICAAIC 2023)*, pp. 680–686.

C12. Eswarapu S.T., et al., **Raj E.D.*** "Integrated Customer Analytics using Explainability and AutoML for Telecommunications." *ICAAIC 2023*, pp. 228–235.

C13. Bhagya J., Thomas J., **Raj E.D.*** "Exploring Explainability and Transparency in Deep Neural Networks: A Comparative Approach." *7th Int. Conf. on Intelligent Computing and Control Systems (ICICCS 2023)*, pp. 664–669.

C14. Shahad P., **Raj E.D.*** "Challenges in Streaming Data Analysis for Building an Adaptive Model for Handling Concept Drifts." *2021 Int. Conf. on System Computation, Automation and Networking (ICSCAN 2021)*, IEEE. **[Best Paper Award]**

C15. Joshi V.S., **Raj E.D.*** "FYEO: A Character Level Model for Lip Reading." *8th Int. Conf. on Smart Computing and Communications (ICSCC 2021)*, pp. 257–262, IEEE. **[Best Paper Award]**

C16. Sreedhar R., Thomas J., **Raj E.D.*** "Detection of Chest Catheters Using Mask R-CNN." *2021 Int. Conf. on Smart Generation Computing, Communication and Networking (Smart Gencon 2021).*

C17. Samariya D., et al., Dhinesh Babu L.D., **Raj E.D.** "A Hybrid Approach for Big Data Analysis of Cricket Fan Sentiments in Twitter." *Int. Conf. on ICT for Sustainable Development (ICT4SD 2015)*, *Advances in Intelligent Systems and Computing*, Vol. 408, Springer, 2016, pp. 503–512.

C18. **Raj E.D.***, Babu L.D.D., Ariwa E. "A Fuzzy Approach to Centrality and Prestige in Online Social Networks." *Proc. Int. Conf. on Information and Analytics (ICIA)*, ACM, 2016.

C19. **Raj E.D.***, Babu L.D.D. "A Model Fuzzy Inference System for Online Social Network Analysis." *2015 Int. Conf. on Computing and Network Communications (CoCoNet 2015)*, IEEE, pp. 582–588.

C20. Sharon Moses J., Babu L.D.D., **Raj E.D.** "Evaluation of Similarity Based Recommender using Fuzzy Inference System." *CoCoNet 2015*, IEEE, pp. 576–581.

C21. **Raj E.D.***, Nivash J.P., Nirmala M., Babu L.D.D. "A Scalable Cloud Computing Deployment Framework for Efficient MapReduce Operations using Apache YARN." *2nd Int. Conf. on Information Technology (ICICES 2014)*, IEEE. *[Scopus indexed]*

C22. Nivash J.P., **Deni Raj E.**, Babu L.D.D., Nirmala M. "A Neural Network Based Framework for Apache YARN." *ICICES 2014*, IEEE. DOI: 10.1109/ICICES.2014.7033783. *[Scopus indexed]*

PhD Students Supervised

#	Scholar Name	Research Area	From	Status / Outcome
1	Akarsh K. Nair	Federated Learning, UAV Security, 6G Communication	2019	**Awarded**;Researcher, Loughborough Univ., UK
2	Ashly Ann Jo	Explainable AI, LLMs, NLP, Transformer Models	2020	Ongoing
3	Shahad P.	Concept Drift, Streaming Data Analysis, IoT	2020	Ongoing
4	Remya Raveendran	Deep Learning, Federated ML, XAI	2022	Ongoing (External)
5	Sasikumar M.	XAI for Time Series, Financial AI	2022	Ongoing

6	*Jeena Thomas*	*Computer Vision and XAI*	2019	**Awarded**; Faculty, IIIT Kottayam

M.Tech. / B.Tech. Projects Guided

Programme	Institution	Approx. Period	Approx. Count
M.Tech. (AI & Data Science)	IIIT Kottayam	2022–Present	25+ projects
B.Tech. (CSE)	IIIT Kottayam	2017–Present	45+ projects
B.Tech. (CSE)	UKF College / LBS Inst.	2012–2017	10+ projects

Project topics include: XAI applications, disease detection using deep learning, federated learning systems, NLP, computer vision, and IoT systems.

Honours, Awards & Recognition

- ★ **ACM Distinguished Speaker / ACM India Eminent Speaker** – Globally recognised invitation-only program; invited to deliver talks at premier institutions under the ACM Distinguished Speakers Program (DSP). Active from 2020.
- ★ **Best PhD Thesis Guide Award (2025)** – Awarded by Indian Institute of Technology Roorkee (IIT Roorkee) for excellence in doctoral supervision.
- ★ **Best Paper Award (2026)** – "Explainable AI for Emotion Divergence : Artist-Intended vs Audience-Perceived Emotions" – Fourth International Conference on Speech Language Technologies for Low Resource Languages, SPringer.
- ★ **Best Paper Award (2025)** – "Transparency in Legal AI: Pioneering Explainable NLP for Enhanced Judgement Prediction" – ACM MIT Bengaluru (Scopus indexed), in collaboration with Linnaeus University, Sweden and Univ. of Applied Sciences, Switzerland.
- ★ **Best Paper Award (2023)** – "Tabular Data Interpretation on Covid-19 Through Explainable Machine Learning" – ETTIS-2023, jointly organised by CDAC Noida, University of Ploiesti (Romania), and Univ. of Haute-Alsace (France).
- ★ **Best Paper Award (2022)** – "Deep Generative Models Under GAN: Variants, Applications and Privacy Issues" – 7th Int. Conf. on Information System Design and Intelligent Applications (INDIA-2022).
- ★ **Best Paper Award (2021 – IEEE ICSCC)** – "FYEO: A Character Level Model for Lip Reading" – 8th Int. Conf. on Smart Computing and Communications, IEEE.
- ★ **Best Paper Award (2021 – IEEE ICSCAN)** – "Challenges in Streaming Data Analysis for Building an Adaptive Model for Handling Concept Drifts" – Int. Conf. on System, Computation, Automation and Networking, IEEE.
- ★ **Best Paper Award (2021 – ICInPro)** – "Effectual Single Image Dehazing with Color Correction Transform and Dark Channel Prior" – 16th Int. Conf. on Information Processing, Bengaluru.
- ★ **Best Ph.D. Thesis Award (2016)** – Awarded by Computer Society of India (CSI).

- ★ **Research Publication Award with Cash Prize (2016)** – VIT University, for outstanding research output.
- ★ **Research Publication Award with Cash Prize (2015)** – VIT University.
- ★ **Distinguished Jury Member – AI Gamechanger 2022** – NASSCOM & MeitY; initiative to accelerate India's AI excellence.
- ★ **Jury Member – Aegis Graham Bell Awards (AGBA) 12th Edition (2022)** – MeitY, NITI Aayog, Skill India, NIC, Maharashtra State Innovation Society.
- ★ **Jury Member – Aegis Graham Bell Awards (AGBA) 11th Edition (2021)** – Same authorities as above; Innovation in Data Science category.
- ★ **Interview Panelist – ML Subject Expert (2023)** – Centre for Memory Studies, IIT Madras; recruitment of research staff.
- ★ **Top 25 Programmers – IEEE Extreme Programmers League / IXPL (2011)** – IEEE, VIT University.
- ★ **Selected by Alcatel-Lucent (Bell Labs, USA)** for Project Internship through a competitive national selection process (2011).
- ★ **Selected by Cognizant Technology Solutions (CTS)** in campus placement (2011).
- ★ **Research featured in international science media:** *ScienceDaily*, *Phys.org*, *ScienceBlog.com*, *Electronic Component News (ECN)*, *ScientificComputing.com*.

Administrative & Institutional Responsibilities

Role	Body / Committee	Period
Associate Dean, Academic Affairs	IIIT Kottayam	2022 – Present
Head of Department, CSE	IIIT Kottayam	(Prior tenure)
Faculty Sponsor – IIIT Kottayam ACM Student Chapter	ACM	2017-2020
IEEE Conference Chair	IEEE ICITIIT 2021 (accept. rate: 30%)	Feb 2021
Secretary	Board of Studies, IIIT Kottayam	2021 – Present
Member	Academic Senate, IIIT Kottayam	2019 – Present
Faculty In-charge, Academic Affairs	IIIT Kottayam	2019 – Present
Faculty In-charge, Admissions (UG)	IIIT Kottayam	2017 – Present
Chairperson	Disciplinary Action Committee	2017 – Present
Technical Chair	ICITIIT 2020, IIIT Kottayam	2020
Assistant Warden	IIIT Kottayam Hostels	(Prior)
Time Table Coordinator	IIIT Kottayam	(Prior)
TPC Member	CICT 2017, ABV IIITM Gwalior	2017
Staff Advisor, 2016 Batch, UKF College of Engg. & Technology	2016–17	
PhD Thesis Examiner	VIT University	Ongoing
PhD Thesis Examiner	Kerala Technological University (KTU)	Ongoing
PhD Thesis Examiner	Anna University	Ongoing

Professional Memberships

Organisation	Category	Since
Association for the Advancement of Artificial Intelligence (AAAI)	Member	2024
ACM SIGCSE	Member	2023
IEEE	Senior Member	2022
IEEE Computer Society	Member	2021
Association for Computing Machinery (ACM) – ACM Distinguished Speaker	Member (DSP)	2015 / 2020
Computer Society of India (CSI)	Member	2015

Invited Talks, Keynotes & Resource Person Roles

Sl.	Talk / Event Title & Venue	Organiser	Year
1	**R**esponsible AI and Block Chain Technology	ESF–IITM Workshop on Distributed Ledgers with Hedera(Talk shared with Prof John Augstine of IIT Madras)	May 2025
2	**R**esponsible AI in the age of LLMs for Healthcare Applications	Digital University of Kerala (DUK)	June 2025
3	**Keynote:** Responsible AI in the era of Data Science: Seeing the story beyond the Numbers	SB College . , Chganganaserry	Feb 2025
4	**E**xplainability Meets MLOps: A Practical Framework for Responsible AI in Industry	Devops Malayalam- DMCON 2025	Oct 2025
5	**Keynote:** From Models to Morals: Engineering Accountability in AI-Powered Technologies	Baseilus Mathews . College, Kayankulam	Aug 2025
6	**Keynote:**Transforming Weak Prompts into Strong, Responsible, and Verifiable Prompts	Workshop on Responsible AI Practices in Higher Education by Mozilla Foundation at Tinkerhup space Kochi	Nov 2025
7	**Keynote:**AI as a Learning Partner: Rethinking Pedagogy for a Gen Z Future	U4U Seminar Organized by HSE University, Russia(QS Ranking 305)	Nov 2025
8	**P**rompt Engineering Prompting, Context Engineering and Verbalized Sampling	Indian Institute of Mass Communication	Nov 2025
9	**F**uture of Data Science	Disha, Adoor, Pathanamthitta District	Sep 2025
10	**U**nderstanding Classification and Clustering Hands-on with Teachable Machine and Orange Data Mining	Digital Humanities Workshop at NIT Calicut	Oct 2025
11	**F**rom You to Algorithm: How Machines Begin to Understand	CHORDS, CSH Department, IIIT Kottayam	June 2025
12	**Keynote:** Explainable and AUDITABLE Machine Learning	ACM Research Opportunity in Computer Science (ACM-ROCS), Organized by ACM	Aug 2024
13	**S**OCIAL network ANALYSIS: Theory and practice	Network Science Research Group, IIIT Kottayam	June 2024
14	**R**esponsible and safe AI	St.George College Aruvithara	Aug 2024

Sl.	Talk / Event Title & Venue	Organiser	Year
15	**Keynote:** Spoke at the International Conference on Memory,Security, and Sustainability	Indian Network for Memory Studies (INMS) and the Centre for Memory Studies (CMS), IIT Madras.	Sep 2024
16	**A**I: Facts, Fiction, Myths, Predictive analytics and the Future	IIMC Kottayam	November 2024
17	**W**orkshop on : AI: Facts, Myths, Predictive Analytics and the Future	IIT Madras	November 2024
18	**R**esponsible and safe AI	St.George College Aruvithara	Aug 2024
19	**A**I for Social Media Monitoring	TechInt (Advanced Training for Police Personnel of State police special branch wing of Kerala Police	December 2024
20	**R**esponsible and safe AI	St.George College Aruvithara	Aug 2024
21	**Keynote:** Responsible AI in the age of LLMs for Precision Medicine and Sustainable Smart City Development	International Conference on Precision Medicine and Sustainable Smart City Development supported by National Science and Technology Council, Taiwan, SRM University	Sep 2024
22	**Keynote:** National Conf. on Research in Emerging Areas (NACORE-24)	Amal Jyothi Engg. College, Kanjirapally	Apr 2024
23	**Keynote:** ATAL FDP on Explainable AI	Adishankara Inst. of Engg. & Technology	Apr 2024
24	Explainable and Interpretable ML – FDP Lecture Series	Holy Grace College of Engg.	Apr 2023
25	Explainable and Interpretable ML – FDP	Amal Jyothi Engg. College, Kanjirapally	Mar 2023
26	Theory and Practice of Social Network Analysis – Workshop for M.Sc. Comp. Social Sciences & Digital Humanities	IIT Jodhpur	Apr 2023
27	Data Storytelling through Data Visualization – Winter School on Computational Methods for Social Sciences	IIT Jodhpur	Dec 2022
28	Machine Learning Design Patterns	IEEE Malabar Section	Aug 2022
29	Clustering and Classification – FDTP on "CS8091 Big Data Analytics" (6-day)	PSG Inst. of Technology & Applied Research, Coimbatore	Jul 2022
30	Research Methodology for Working Professionals	Hindustan Petroleum Corporation Ltd (HPCL)	Jan 2022
31	AI and Data Science – Online Public Talk, 1300+ registrations	Malayala Manorama (Popular Print Media)	Jan 2022
32	Berchmans Lecture Series in Mathematics – Invited Talk	SB College, Changanacherry	Jan 2022
33	AI and Data Science to IAS Aspirants	SSKBRC, Erattupetta	Sep 2023
34	**ACM Invited Talk:** Pursuing a Career in Data Science	Velagapudi Ramakrishna Siddhartha Engg. College (VRSEC), Vijayawada	Jun 2021

Sl.	Talk / Event Title & Venue	Organiser	Year
35	Opportunities in Artificial Intelligence and Data Science (Alan Turing Day Celebration)	St. George's College, Aruvithara	Jun 2021
36	Machine Learning Design Patterns – KTU FDP (5-day)	ToCH College of Engg. & Technology	Jul 2021
37	Research Trends in Data Science – KTU FDP (5-day)	St. Joseph College of Engg., Pala	Jul 2021
38	Research challenges & opportunities in Online Social Networks – KTU FDP (5-day)	SCMS School of Engg. & Technology	May 2021
39	**Keynote Speaker** – ICIAETM-2021	Int. Conf. on Innovative Applications of Emerging Technologies (online, ≈200 participants)	Apr 2021
40	Machine Learning Research Concerns and Challenges – STTP (6-day, ≈350 participants)	Amal Jyothi College of Engg.	Aug 2021
41	Sustainability and Big Data – AICTE-sponsored STTP (5-day)	College of Engg., Karunagapally	Mar 2021
42	DATA SCIENCE – Process, Hype and Future (≈100 participants)	Kristu Jayanthi College, Bangalore	Jun 2020
43	Data Science and Artificial Intelligence	St. Gitts College of Engg.	Oct 2019
44	Trends in Data Science and Artificial Intelligence	ToCH College of Engg. & Technology	Feb 2020
45	Data Science and Data Engineering – DST-NIMAT Workshop	Cloud Solutions for Emerging IoT Applications	Dec 2018
46	Social Big Data	Sree Ramakrishna Engg. College, Coimbatore	June 2017

Conferences, Workshops & Seminars Organised

1. **Organizer** – 5-Day bootcamp on Responsible Computing, 2025. ; 17-21 March,2025.
2. **IEEE Conference Chair** – 2nd Int. Conf. on Innovative Trends in Information Technology (ICITIIT'21), Feb 11–12, 2021. Virtual conference; paper acceptance rate: 30%.
3. **Technical Chair** – ICITIIT 2020, IIIT Kottayam.
4. **One-Day National Training** on Data Science and Machine Learning: A Learning Journey – July 28, 2021.
5. **Workshop on Design Thinking** by Mr. Jim J. Seelan (TCS Innovation), IIIT Kottayam.
6. **One-Day Workshop on Internet of Things** – IIIT Kottayam, January 13, 2017.
7. **Webinar:** "Through and Beyond COVID-19: Integrating Social Responsiveness and Technology" – June 23, 2020. Resource Persons: Mr. Rejin Narayan & Mr. Jim J. Seelan (TCS).
8. **Webinar:** "Industry 4.0: Digital Manufacturing" – Resource Person: Mr. Thadeus Devasagayam, Consultant–Digital Manufacturing.
9. **Webinar:** "IT Industry Expectation from Engineering Students" – Resource Person: Ms. Rekha Mathews (Leader-HR, Nissan Digital, Nissan Motor Corporation).
10. **Guest Lecture:** "Software Job – What? Why? How? What If?" by Mr. Subin Kuttapan (Senior Software Engineer, Oracle Corporation), IIIT Kottayam.

Innovation in Teaching Methodology

1. Designed and operationalised large-scale POGIL classroom workflows, demonstrating scalability from mid-size (65 students) to large cohorts (130 students) through structured roles, guided worksheets, and collaborative assessment strategies.
2. Created *rstudio.iiitkottayam.ac.in* – Interactive R programming server for postgraduate students; hosted at IIIT Kottayam campus. Enables real-time coding and execution for M.Tech. students without local setup.
3. Created *rconnect.iiitkottayam.ac.in* – Platform enabling students to publish R/Shiny and Flask data dashboards and project outputs. Hosted at IIIT Kottayam campus.
4. Integrated **Virtual Programming Lab** in Moodle for the Data Structures & Programming course, enabling automated test-case-based evaluation and improving coding proficiency.
5. Introduced **J-FLAP simulations** in the Theory of Computation course for interactive visualisation of automata and Turing machines.
6. Worked with the IIIT-K IT team to develop a custom **Open-Source Online Learning Platform** (Big Blue Button + Moodle integration); institute received **appreciation from the Ministry of Education** (October 2020) for this initiative.
7. Introduced **real-life industrial case studies** in DSC513 (Introduction to AI & Data Science) to bridge academic content with industry practice.
8. Added **Virtual Programming Lab module** in Moodle for better programming instruction and ease of evaluation.
9. Worked with the IT team to develop the **institute's result-publishing portal**, which has been in active use since deployment.

Review & Editorial Activities

Journal Reviewer

- IEEE Transactions on Emerging Topics in Computing
- Applied Soft Computing (Elsevier)
- Information Technology and People
- Physics Letters A
- Journal on Wireless Communications and Networking
- International Journal of Data Science and Analytics

Conference Technical Programme Committee / Reviewer

- Reviewer – IEEE 2nd Int. Conf. on Anti-Cyber Crimes (ICACC) – 2016
- TPC Member – Conference on Information and Communication Technology (CICT) 2017, ABV IIITM Gwalior

PhD Thesis Examination

- PhD Thesis Examiner – Vellore Institute of Technology (VIT)
- PhD Thesis Examiner – Kerala Technological University (KTU)

Technical Skills

Category	Tools / Technologies
Programming Languages	Python, R, Java, MATLAB, C, C++, Bash
ML / DL Frameworks	TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers
XAI Tools	SHAP, LIME, Grad-CAM, Captum, Integrated Gradients, DeepLIFT
Cloud / Big Data	AWS, Microsoft Azure, Apache Spark, Hadoop/YARN, Apache Storm
Teaching / LMS Platforms	Moodle, Big Blue Button, RStudio Server, Shiny, Flask
Version Control / DevOps	Git, GitHub (raj007iiit), Docker
Document Preparation	LaTeX, Microsoft Office Suite
Gen AI Tools	Claude.ai, Perplexity, Grok, Napkin.ai

Declaration

I hereby declare that the information furnished above is true, complete, and correct to the best of my knowledge and belief. No part of it is false, and nothing has been concealed or suppressed.

Place: Kottayam, Kerala
Date: 27/02/2026

Signature: Ebin Deni Raj
(Dr. Ebin Deni Raj)

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